

Data Structures and Algorithms - Assignment 3

1. Implement following algorithms for multi digit operands (use space to separate operands)

- postfix evaluation
- prefix evaluation

2. <https://leetcode.com/problems/valid-parentheses>

3. <https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string>

4. <https://leetcode.com/problems/palindrome-linked-list>

5. Leetcode Problem : <https://leetcode.com/problems/middle-of-the-linked-list/description>

6. Leetcode Problem : <https://leetcode.com/problems/delete-the-middle-node-of-a-linked-list/description>

7. Leetcode Problem : <https://leetcode.com/problems/linked-list-cycle/description>

Optional

8. Write program to implement linear queue in which front and rear starts at 0.

9. Write a program to implement descending stack. (Initialize top = SIZE)

10. Implement circular queue using counter method (maintain count of elements in a queue and use it for checking empty and full conditions)

11. Paperwork: Convert following examples from Infix to Prefix and Postfix.

- $K + L - M * N + (O \$ P) * W / U /$
- $V * T + Q (A + B) * C - (D - E) * (F + G)$

12. Paperwork: Convert following expression to Postfix and Prefix using stack.

- $5+9-4*(8-6/2)+1$(7-3)$